UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment 4

CLO Mapping: CLO3

Course Code	MAT233	Semester	Spring 2026
Course Title	Linear Algebra		
Instructor	Adnan Rafiq Siddiqui		
Total Marks	20		

Sr.	Name of student	Registration no.	Section	Assignment marks	Total marks
1					

Question No. 1.

Any $\mathbf{2} \times \mathbf{2}$ image block can be represented as a vector in the vector space $M_{2\times 2}$. First show that the set $S = \{A_1, A_2, A_3, A_4\}$ is a basis (Linearly independent and Spans) for M_{22}, then express A as a linear combination of the vectors in S (find all constant multiples of A_1, A_2, A_3, A_4).

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 6 & -2 \\ 5 & 3 \end{bmatrix}$$

Question No. 2.

The matrix A is defined as:

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{-3} & \mathbf{1} & \mathbf{-1} \\ \mathbf{-7} & \mathbf{5} & \mathbf{-1} \\ \mathbf{-6} & \mathbf{6} & \mathbf{-2} \end{bmatrix}$$

Find the Eigenvalues and corresponding Eigenvectors of the matrix A.

A 3×3 matrix typically yields three linearly independent eigenvectors. If you only find two, explain the mathematical reason behind this shortage.